

Project Helios

Part 3: Project Scheduling, Resources & Quality

Mercury-based autonomous manufacturing for Dyson swarm construction

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1. Gantt Chart and Critical Path Analysis

1.1 Project Timeline Visualization

The Gantt chart below illustrates the complete project timeline for Project Helios, spanning from Year 1 to Year 8 of the 20-year mission. The chart displays all major work packages derived from the Work Breakdown Structure in Part 2, showing task dependencies, durations, and the critical path.

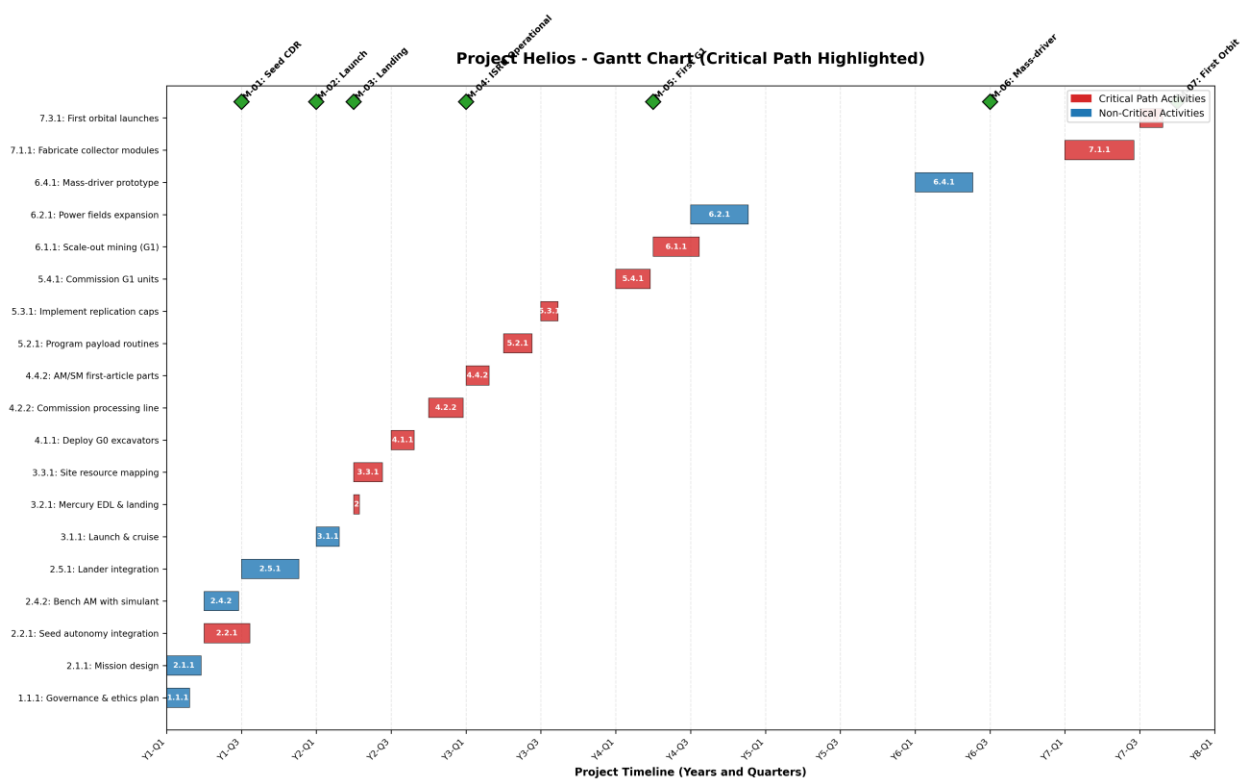


Figure 1: Project Helios Gantt Chart with Critical Path

1.2 Critical Path Analysis

The critical path represents the longest sequence of dependent activities that determines the minimum project duration. Any delay in critical path activities will directly impact the project completion date.

Critical Path Sequence:

- 2.2.1: Seed autonomy stack integration (16 weeks)
- 3.2.1: Mercury EDL & landing (2 weeks)
- 3.3.1: Site resource mapping (10 weeks)
- 4.1.1: Deploy G0 excavators (8 weeks)
- 4.2.2: Commission processing line (12 weeks)
- 4.4.2: AM/SM cell first-article parts (8 weeks)
- 5.2.1: Program payload-centric routines (10 weeks)
- 5.3.1: Implement replication caps & QA (6 weeks)
- 5.4.1: Commission first G1 units (12 weeks)
- 6.1.1: Scale-out mining (G1 fleet) (16 weeks)
- 7.1.1: Fabricate collector modules (24 weeks)
- 7.3.1: First orbital launches (8 weeks)

Rationale: This critical path aligns with the technical architecture where no mass production or orbital launches can occur until the complete chain of autonomy integration, ISRU establishment, first-article fabrication, replication governance, and fleet scale-out are operational. This sequence directly reflects the Mercury deployment strategy and phased bootstrap approach outlined in Parts 1 and 2.

2. Human Resources Requirements

The human resources requirements are organized into five time buckets spanning the 20-year project timeline. Resource allocation peaks are indicated for each role based on the project phases and technical demands.

Time Buckets:

TB1: Years 1-4 (Seed Deployment & Bootstrap)

TB2: Years 5-8 (Infrastructure Expansion)

TB3: Years 9-12 (Module Production Ramp-up)

TB4: Years 13-16 (Full-Scale Operations)

TB5: Years 17-20 (Mature Operations & Sustainability)

Role / Skill	TB1 (Y1-Y4)	TB2 (Y5-Y8)	TB3 (Y9-Y12)	TB4 (Y13-Y16)	TB5 (Y17-Y20)
Project Manager (PMP)	X	X	X	X	X
Chief Systems Engineer	PEAK	PEAK	X	X	X
Autonomy & Robotics Lead	PEAK	PEAK	X	X	X
ISRU / Chemical Process Engineer	X	PEAK	X	X	X
Manufacturing (AM/SM) Engineer	X	PEAK	PEAK	X	X
Test & QA Lead	X	X	PEAK	X	X
Safety & Ethics Officer	X	X	X	X	X
Launch & EDL Operations Lead	X	X	X	X	X
EM Launch / Mass-Driver Engineer	-	X	PEAK	X	-
Orbital Ops & Tracking Lead	-	-	X	PEAK	X
Supply Chain & Logistics	X	PEAK	PEAK	X	X

Manager					
Radiation / Materials Scientist	X	X	X	X	X

Note: PEAK indicates periods of maximum resource demand for that role. X indicates steady-state requirements. Dash (-) indicates role not required in that period.

2.1 Resource Allocation Rationale

Early Phase Focus: Heavy emphasis on autonomy, ISRU, AM/SM, and governance during seed deployment and bootstrap phases aligns with establishing core capabilities.

Mass-Driver Engineering: Peaks during infrastructure expansion when the prototype is built and tested, then reduces as operations stabilize.

Orbital Operations: Ramps up significantly once collector module deployment begins in Years 7-12 and requires sustained support through aggregation phases.

Systems Engineering: Maintains high levels early for integration and V&V, then transitions to oversight and optimization in later phases.

3. Project Roles and Responsibilities

3.1 Key Project Roles

The following individuals hold primary accountability for project success:

Sponsor: Dr. Aria Novik

Project Manager: Yatharthha Kaushal

Chief Systems Engineer: Mina Okoye

Autonomy Lead: Leo Zhang

ISRU Lead: Sofia Martínez

Manufacturing Lead (AM/SM): Hiro Tanaka

Test & QA Lead: Priya Nair

Safety & Ethics Officer: Omar Haddad

Launch Operations Lead: Nora Kaczmarek

Orbital Operations Lead: Ethan Rossi

Supply Chain Lead: Jules Mbaye

Communications & Stakeholder Lead: Ana Popescu

3.2 RACI Matrix

The RACI matrix defines responsibilities across major work packages:

Legend: R = Responsible (does the work) | A = Accountable (ultimate ownership) | C = Consulted (provides input) | I = Informed (kept updated)

WP / Activity	Sponsor	PM	SysEng	Auto	ISRU	Mfg	QA	Safety	Launch	Orbital	Supply	Comms
1.1 Governance & ethics	A	C	C	I	I	I	C	R	I	I	I	I
2.2 Seed autonomy integration	I	A	C	R	I	I	C	I	I	I	I	I

3.2 EDL & landing ops	I	A	C	C	I	I	C	I	R	I	I	I
3.3 Site survey & mapping	I	A	C	R	R	I	C	I	I	I	I	I
4.x ISRU commissioning	I	A	C	I	R	C	C	I	I	I	C	I
4.4 AM/SM first-articles	I	A	C	I	C	R	C	I	I	I	C	I
5.x Replication & caps	I	A	C	R	C	C	R	R	I	I	I	I
6.4 Mass-driver prototype	I	A	C	C	C	R	C	I	R	I	C	I
7.1 Module fabrication	I	A	C	C	C	R	C	I	I	I	C	I
7.3 Orbital launches	I	A	C	C	I	C	C	I	R	R	C	I

Note: Work packages reflect the WBS hierarchy from Part 2. Responsibilities emphasize payload-centric autonomy, replication governance, and the ISRU/AM manufacturing flow consistent with the technical architecture.

4. Quality Management

4.1 Quality Standards

Project Helios implements the following quality standards tailored to the unique requirements of autonomous space manufacturing:

Design & Development Gates: Formal reviews at each major milestone including Critical Design Review (CDR) for the seed payload and Operational Readiness Reviews at each phase gate. Acceptance criteria include $\geq 95\%$ subsystems online post-landing, documented ISRU feedstock rates, replication QA pass rates, mass-driver fire acceptance, and tracking of first 10 orbital modules.

Replication Governance Standard: Mandatory hard-coded generation caps, authenticated kill-switch mechanisms, resource and energy quotas per unit, and auditable logs of all replication events. Zero tolerance policy for uncontrolled replication.

Environmental & Reliability Standard: All systems must meet Mercury environmental extremes (-170°C to $+430^{\circ}\text{C}$ exposure ranges). Requirements include radiation-hard electronics, thermal protection systems, dust mitigation protocols, modular redundancy, and graceful degradation policies.

Manufacturing & Inspection Standard: First-article inspection for all AM/SM parts fabricated from regolith-derived materials. In-process controls including dimensional verification, non-destructive evaluation, and process qualification before volume production authorization.

Verification & Validation (V&V): Hardware-in-the-loop (HIL) testing for G1 units, end-to-end simulation of EDL sequences, ISRU processing chains, replication routines, mass-driver operations, and orbital deployment scenarios.

Ethics & Compliance Standard: Adherence to international space law, planetary protection protocols, and UN/COSPAR guidance. Transparent governance with external review mechanisms and public reporting of replication activities.

4.2 Quality Assurance vs. Quality Control

Project Helios distinguishes between process-focused Quality Assurance and product-focused Quality Control:

Quality Assurance (QA) - Process-Focused:

- Phase-gate reviews (CDR, ORR) with documented entry/exit criteria
- Configuration control across autonomy, ISRU, and AM toolchains
- Supplier qualification and incoming assurance for imported avionics
- Failure Mode and Effects Analysis (FMEA) and hazard analyses on replication processes
- Code audits using formal methods for autonomy stack and kill-switch logic

Quality Control (QC) - Product-Focused:

- Incoming inspection of imported electronics and critical components
- First-article inspection of all printed and processed parts
- In-process sensors for AM monitoring (temperature, energy, layer metrics)
- Functional testing for daughter units (G1) before fleet commissioning
- Acceptance testing for mass-driver firings and orbital module telemetry

4.3 Quality Roles and Responsibilities

Quality management responsibilities are distributed across the project team:

Test & QA Lead (Priya Nair): Accountable for overall QA/QC planning, audit execution, and conformance reporting.

Manufacturing Lead (Hiro Tanaka): Responsible for AM/SM process control, first-article inspections, and manufacturing quality.

Autonomy Lead (Leo Zhang): Responsible for software validation, HIL testing, and safe-mode behavior verification.

Safety & Ethics Officer (Omar Haddad): Accountable for replication governance compliance and kill-switch verification protocols.

Chief Systems Engineer (Mina Okoye): Accountable for system-level V&V integration and technical performance measures.

Project Manager (Yatharthha Kaushal): Accountable for phase-gate approvals and Quality Management System (QMS) enforcement.

5. Conclusion

Part 3 provides the project scheduling, resource allocation, and quality framework necessary to execute Project Helios successfully. The Gantt chart and critical path analysis demonstrate the logical sequencing of activities from seed deployment through bootstrap replication to full-scale orbital module production.

The human resources plan ensures appropriate expertise is available when needed, with peak allocations aligned to technical demands in each phase. The RACI matrix clarifies decision-making authority and communication pathways across the distributed project team.

The quality management framework emphasizes both process discipline (QA) and product verification (QC), with special attention to the unique challenges of autonomous replication, extreme environmental conditions, and ethical governance. This comprehensive approach positions Project Helios to achieve its ambitious goal of establishing self-sustaining manufacturing infrastructure on Mercury for Dyson swarm construction.

Work Cited

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